# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

When $CH_3COCH_3$ reacts with $Cl_2$ and $NaOH$, which of the following is formed

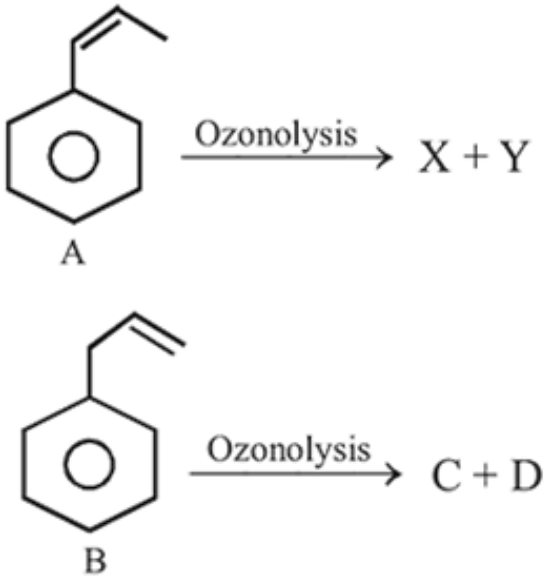
**(A)** $CHCl_3$

**(B)** $CCl_4$

**(C)** $CCl_2H_2$

**(D)** $CH_3Cl$

**Q.2**




From the ozonolysis products, the two isomers A and B can be distinguished with the help of

**(A)** Fehling solution

**(B)** Tollen's reagent

**(C)** Haloform test

**(D)** only spectroscopy

**Q.3**

Which is major product formed when acetone is heated with iodine and potassium hydroxide?

**(A)** Iodoacetone

**(B)** Acetic acid

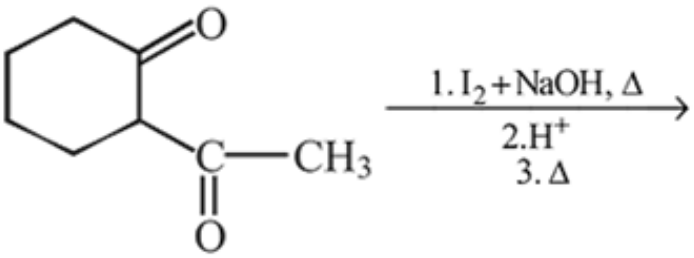
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**(C)** Iodoform

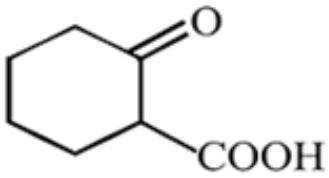
**(D)** Acetophenone

**Q.4**

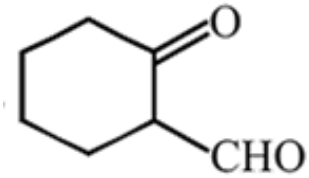
End product of the following sequence of reactions are:



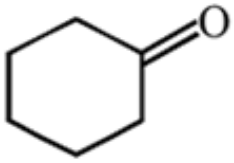

**(A)** yellow ppt. of $CHI_3$,



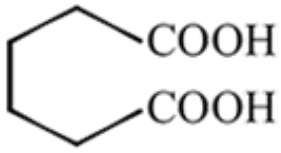

**(B)** yellow ppt. of $CHI_3$,




**(C)** yellow ppt. of $CHI_3$,




**(D)** yellow ppt. of $CHI_3$,




**Q.5**

Formaldehyde can be distinguished from acetaldehyde by the use of:

**(A)** Schiff's reagent

**(B)** Tollen's reagent

**(C)** $I_2$ /Alkali

**(D)** Fehling's solution

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**Q.6**

Compound 'A' (molecular formula $C_3H_8O$) is treated with acidified potassium dichromate to form a product ' B' (molecular formula $C_3H_6O$). 'B' forms a shining silver mirror on warming with ammonical silver nitrate. 'B' when treated with an aqueous solution of $H_2NCONHNH_2.HCl$ and sodium acetate gives a product ' C '. Identify the structure of ' $C'$

**(A)** $CH_3CH_2CH = NNHCONH_2$

**(B)** $CH_3 - \underset{\underset{CH_3}{|}}{C} = NNHCONH_2$

**(C)** $CH_3 - \underset{\underset{CH_3}{|}}{C} = NCONHNH_2$

**(D)** $CH_3CH_2CH = NCONHNH_2$

**Q.7**

An organic compound ' $A$ ' has the molecular formula $C_3H_6O$. It undergoes iodoform test. When staturated with $HCl$ it gives ' $B$ ' of molecular formula $C_9H_{14}O$. ' $A$ ' and ' $B$ ' respectively are

**(A)** Propanal and mesitylene

**(B)** Propanone and mesityl oxide

**(C)** Propanone and 2,6-dimethyl-2, 5-heptadien4 -one

**(D)** Propanone and mesitylene oxide

**Q.8**

Which is not true about acetophenone?

**(A)** It reacts with 2,4 -dinitrophenylhydrazine to form 2,4 -dinitrophenylhydrazone

**(B)** It reacts with Tollen's reagent to form silver mirror

**(C)** It reacts with $I_2/NaOH$ to form iodoform

**(D)** On oxidation with alkaline $KMnO_4$ followed by hydrolysis it gives benzoic acid

**Q.9**

Which of the following is incorrect?

**(A)** $FeCl_3$ is used in detection of phenols.

**(B)** Fehling solution is used in detection of glucose.

**(C)** Tollen's reagent is used in detection of unsaturation.

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**(D)** $NaHSO_3$ is used in detection of carbonyl compounds.

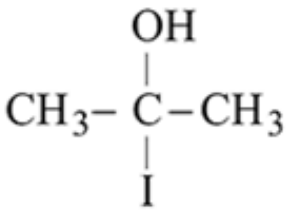
**Q.10**

Consider the following transformations:

$$CH_3COOH \xrightarrow{CaCO_3} A \xrightarrow{heat} B \xrightarrow[NaOH]{I_2} C$$

The molecular formula of $C$ is

**(A)**



**(B)** $ICH_2 - COCH_3$

**(C)** $CHII_3$

**(D)** $CH_3I$

**Q.11**

The most appropriate reagent for the conversion of 2 -pentanone to butanoic acid is

**(A)** sodium hypoiodite

**(B)** $O_2$

**(C)** acidified $KMnO_4$

**(D)** alkaline $KMnO_4$

**Q.12**

The light yellow compound produced when acetone reacts with iodine and alkali, is

**(A)** $CH_3.CO \cdot CH_2I$

**(B)** $CH_3I$

**(C)** $CHI_3$

**(D)** None of these

**Q.13**

When aniline reacts with oil of bitter almonds $(C_6H_5CHO)$ condensation takes place and benza derivative is formed. This is known as

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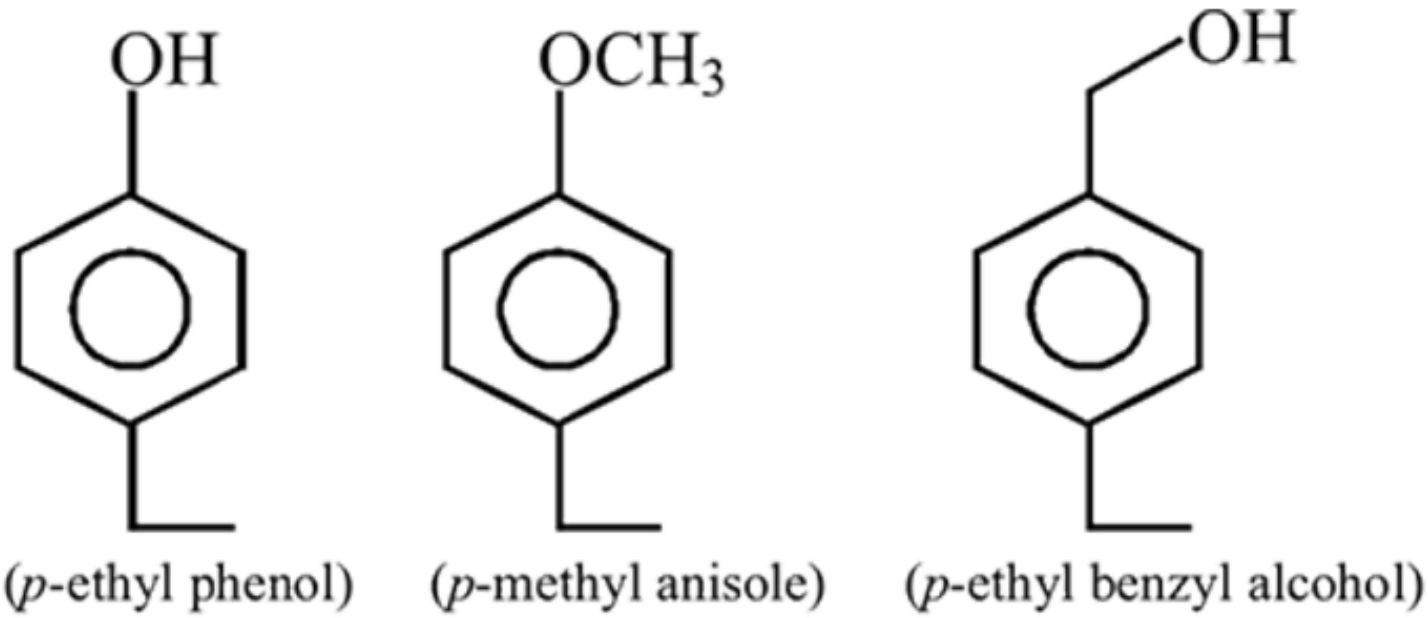
**(A)** Million's base

**(B)** Schiff's reagent

**(C)** Schiff's base

**(D)** Benedict's reagent

**Q.14**




(*p*-ethyl phenol) (*p*-methyl anisole) (*p*-ethyl benzyl alcohol)

Above compounds can be differentiated by using the reagent:

**(A)** $NaOH$, Tollen's reagent, $FeCl_3$

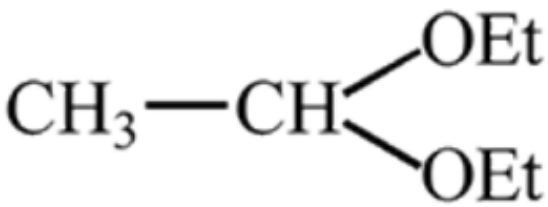
**(B)** $CrO_3$, Tollen's reagent, $FeCl_3$

**(C)** Tollen's reagent, $CrO_3$, $FeCl_3$

**(D)** Na, Tollen's reagent, $FeCl_3$

**Q.15**

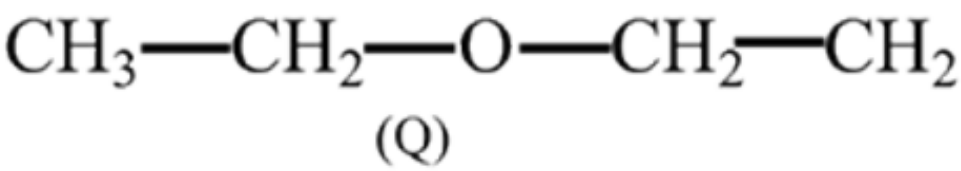
Compound




(P) (Acetal)

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and



(Q)

can be differentiated by:

**(A)** $H_3O^{\oplus}$, Na

**(B)** $H_3O^{\oplus}$, Tollens' test

**(C)** $H_3O^{\oplus}$, Fehling test

**(D)** All of these

## Answers & Solutions

**Q.1 Answer:**

$CHCl_3$

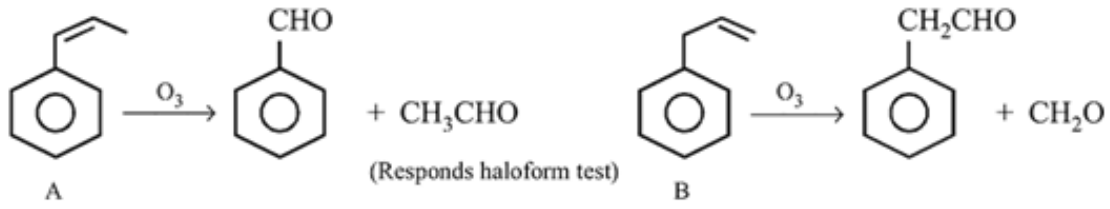
**Solution:**

$$2CH_3COCH_3 + Cl_2/NaOH \rightarrow 2CHCl_3 + 2CH_3COONa + HCl$$

**Q.2 Answer:**

Haloform test

**Solution:**




**Q.3 Answer:**

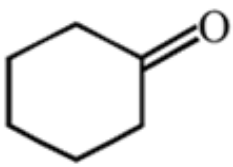
Iodoform

**Solution:**

$$CH_3COCH_3 + 3I_2 + 4KOH \longrightarrow CHI_3 + CH_3COOK + 3HI + 3H_2O$$

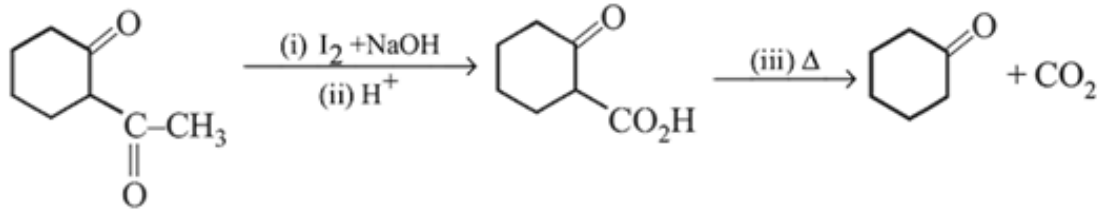
**Q.4 Answer:**

yellow ppt. of $CHI_3$,




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**Solution:**



**Q.5 Answer:**

$I_2$ /Alkali

**Solution:**

**Only acetaldehyde and methyl ketones give iodoform test.**

**Q.6 Answer:**

$CH_3CH_2CH = NNHCONH_2$

**Solution:**

$C_3H_8O \xrightarrow{K_2Cr_2O_7/H^+} C_3H_6O \xrightarrow{H_2NCONHNH_2} C$

**A** **B** $(-CHO)$ **\ Since B reduces Tollen's reagent, it indicates that it has an -CHO group, so it must be** $CH_3CH_2CHO$**. Hence ** $CH_3CH_2CH_2OH \rightarrow$

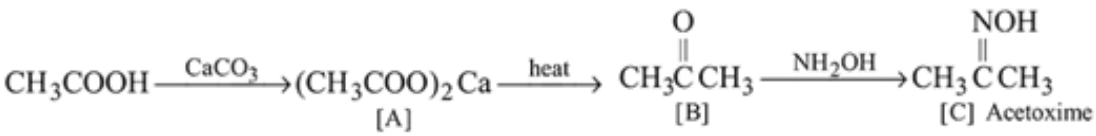
$CH_3CH_2CHO \xrightarrow{H_2NNHCONH_2} CH_3CH_2CH = NNHCONH_2$ **\ [A]** **[B]**

**[C]**

**Q.7 Answer:**

Propanone and mesitylene oxide

**Solution:**



**Q.8 Answer:**

It reacts with Tollen's reagent to form silver mirror

**Solution:**

**Acetophenone** $(C_6H_5COCH_3)$ **being a ketone, does not reduce Tollen's reagent.**

**Q.9 Answer:**

Tollen's reagent is used in detection of unsaturation.

**Solution:**

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**Unsaturation is not detected by Tollen's reagent, a reagent for differentiating between aldehydes and ketones.**

**Q.10 Answer:**

$CHII_3$

**Solution:**

$$CH_3COOH + CaCO_3 \rightarrow \underset{(A)}{(CH_3COO)_2Ca} \xrightarrow{Heat} \underset{(B)}{\begin{matrix}H_3C\\H_3C\end{matrix}>CO} \xrightarrow{I_2+NaOH} \underset{(C)}{CHI_3}$$

**Q.11 Answer:**

sodium hypoiodite

**Solution:**

**Haloform reaction converts methyl ketones to acids directly.**

**Q.12 Answer:**

$CHI_3$

**Solution:**

$CHI_3$ **is yellow compound when iodine reacts with** $NaOH$ **and ketone.**

**Q.13 Answer:**

Schiff's base

**Solution:**

**Benzaldehyde reacts with primary aromatic amine to form schiff's base** $C_6H_5CH = O + C_6H_5NH_2 \longrightarrow C_6H_5CH = NC_6H_5$

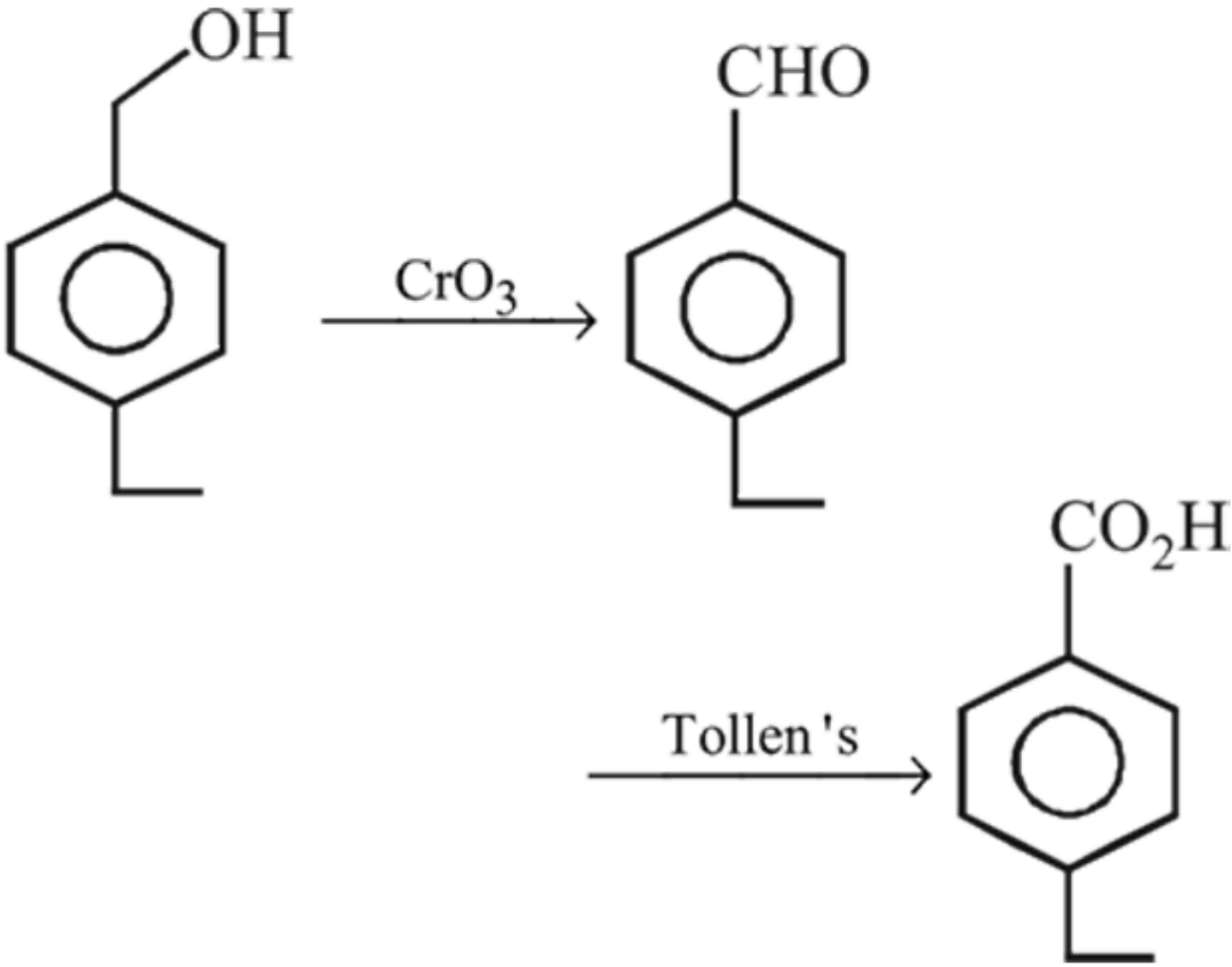
Benzaldehyde Aniline Benzylidene aniline

**Q.14 Answer:**

$CrO_3$, Tollen's reagent, $FeCl_3$

**Solution:**

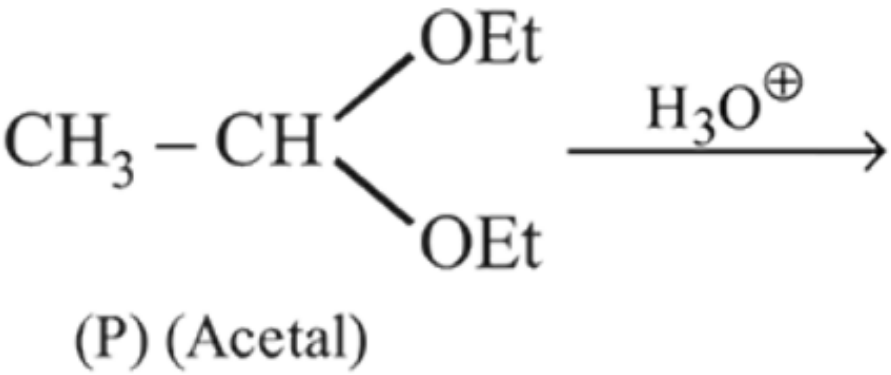
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**Q.15 Answer:**

All of these

**Solution:**




(P) (Acetal)

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$CH_3CHO$; **differentiated by Na, Fehling Tollen's test**

$$CH_3-CH_2-O-CH_2-CH_3$$
(Q)

$$\xrightarrow{H_3O^{\oplus}} EtOH$$

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